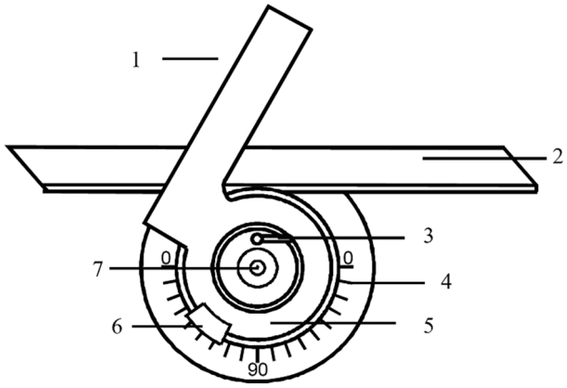


Vernier Bevel Protractor

1. Stock 2. Movable blade
3. Clamp of Blade 4. Main Scale
5. Disc 6. Vernier Scale
7. Locking Screw

Below the disc, blade is fitted and it can be moved lengthwise. The circular disc and blade can be adjusted to measure the angle depends upon the angle of object. Measuring the angle of dead centre of a lathe with the help of vernier Bevel protractor which is shown in figure.

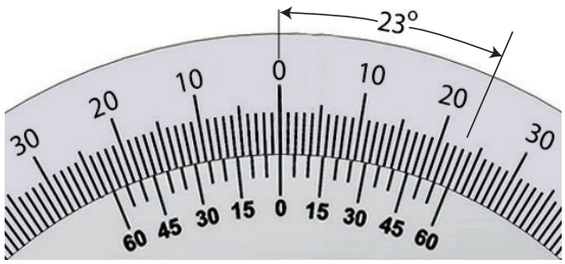
Finding the Least Count of Vernier Bevel Protractor

The value of one division of main scale is one degree. The one side vernier scale has 12 division which is equal to 23^0 in the main scale. The value of one division of vernier scale is $\frac{23^0}{12}$.

Least Count of
Vernier Bevel Protractor = 1 MSD – 1 VSD
LCV Bevel Protractor = One main Scale Division – One Vernier Scale Division

$$= 1 - \frac{23^0}{12}$$

$$= 2 - \frac{23^0}{12}$$



Reading with Vernier Bevel Protractor

(Let us take 1MSD as two Because the 1VSD is greater than 1MSD)

$$= \frac{24 - 23^0}{12}$$

$$= \frac{1^0}{12}$$

$$= \frac{60}{12} \ (1^0 = 60 \text{ minutes})$$

$$= 5 \text{ minutes}$$

3.7 SINE BAR

Measuring instruments like combination set, and vernier bevel protractor are used to measure the angles directly the measuring instrument which is used to measure the angle with the help of sine Formula, so it is called as 'Sine bar'. The least count of Sine bar is one minute.

$$Sin\theta = \frac{Opposite}{Hypotenuse}$$

Sine bar consist of an accurately lapped steel bar and two rollers. The two roller of the same size is fitted with both ends of the sine bar. The size of the sine bar is specified by the distance between the centers of two rollers. Sine bar and slip

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